

Progress Report 1

OneFitPass Canada

CSIS 3275 – 070, Software Engineering

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1. Business Case

1.1 Executive Summary

The Canadian fitness market remains highly fragmented, with users often required to purchase separate memberships for different gyms or chains. This leads to significant inconveniences, including high cumulative costs (typically \$40–\$90 per month per gym in urban areas like Vancouver and Toronto), the need for multiple apps and access cards, difficulties with cancellations, and limited flexibility for individuals who travel within cities, provinces, or work in varying locations (IBISWorld, 2025 Fitness Avenue, 2025).

Canada has approximately 9,395 gym, health, and fitness club businesses as of 2025, generating an industry revenue of \$5.8 billion, with membership penetration rates stabilizing around 15.5–21.1% of adults (IBISWorld, 2025; Fitness Avenue, 2025a; Health & Fitness Association, 2025). While services like ClassPass exist, they primarily focus on group classes, yoga, and boutique studios rather than unlimited or high-frequency access to traditional weight-training gyms (Trustpilot, n.d.; ClassPass, as cited in Yahoo Health, 2025).

1.2 Business Goals and Solutions

1.2.1 Goals

The primary objectives of **OneFitPass** Canada are:

- To provide a convenient, affordable way to access multiple gyms without multiple subscriptions, reducing barriers to consistent fitness and increasing overall physical activity levels.
- To partner with 10–15% of gyms in high-density areas like Metro Vancouver (estimated hundreds of facilities) within the first 12–18 months, expanding to British Columbia, Alberta, and Ontario thereafter.
- To deliver stable supplementary revenue for partner gyms, particularly during low-occupancy periods (mornings, mid-days, evenings), enabling them to better compete with large chains.
- To achieve \$1,000,000 in Annual Recurring Revenue (ARR) within 3–4 years through commissions on visits (25–40%), premium subscription tiers, and value-added services (e.g., integrations with sports nutrition brands).
- To enhance user loyalty by enabling location-agnostic training: switching gyms based on schedule, residence, or travel without financial loss.

1.2.2 Solutions

OneFitPass Canada is a web platform **with a companion mobile app** that functions as a unified digital fitness pass targeted at traditional gym facilities. Users subscribe to a plan (e.g., 8–20 visits per month or unlimited tiers), receive a personal QR code or digital card in the app, and scan it for entry at any partner gym—no extra cards, sign-ups, or per-visit fees required.

For users:

- Substantial savings compared to holding 2–4 separate memberships (average single gym cost \$40–\$90/month in cities like Vancouver/Toronto, with urban premiums 30–50% higher than rural) (Fitness Avenue, 2025b).
- True flexibility: train in different neighborhoods (e.g., Metrotown to Downtown Vancouver to Surrey), test new facilities risk-free, and adapt to life changes.
- Single account, unified app experience, transparent visit history, and easy subscription management.

For partner gyms:

- Increased client flow during underutilized hours without their own marketing investment.
- Pay-per-visit revenue model: platform pays per actual check-in (minus commission), ensuring alignment with real usage.
- Option to offer discounts on OneFitPass to existing members, boosting retention.

High-level implementation: QR/NFC-based check-in system, partner portal for access/schedule management and payouts, secure payment integration. This directly tackles market fragmentation, where users express frustration with multiple subscriptions and limited multi-gym options in current services (Fitness Avenue, 2025a; Trustpilot, n.d.).

2. Problem Statement

2.1 Problem Background: Challenges of Key Personas

The Canadian fitness industry, while valued at \$5.8 billion with approximately 9,395 facilities nationwide, is highly fragmented with users often managing multiple memberships across chains like GoodLife Fitness, Anytime Fitness, and independent gyms (IBISWorld, 2025; Fitness Avenue, 2025a). This fragmentation disproportionately affects specific user groups, or personas, who face daily challenges in maintaining consistent fitness routines.

- **Urban Professional (e.g., 25–45 years old, working in Vancouver or Toronto):** These individuals often commute between home, office, and client sites, requiring gym access

in multiple neighborhoods (e.g., Metrotown to Downtown Vancouver to Surrey). They typically hold 2–3 separate memberships costing \$120–\$240/month, dealing with multiple apps, cancellation hurdles, and wasted fees when life changes disrupt routines. Post-pandemic, 68% of this group report reduced gym attendance due to inflexible options (Health & Fitness Association, 2025).

- **Student or Young Adult (e.g., 18–30 years old, budget-conscious):** Living in high-density areas like Metro Vancouver, they seek variety to try different facilities but are deterred by high entry fees (\$40–\$100 join fees) and rigid contracts. With membership penetration at only 15.5–21.1% among adults, many opt out entirely, citing costs and lack of short-term options as barriers (Fitness Avenue, 2025a).
- **Frequent Traveler or Remote Worker (e.g., digital nomads or inter-province commuters):** These users travel between cities (e.g., Vancouver to Calgary) and need seamless access without re-registering. Current systems force them to pay drop-in rates (\$15–\$25/visit) or maintain unused memberships, leading to frustration and inconsistent habits.

These personas highlight a broader issue: the market's reliance on siloed memberships discourages broader participation, especially in urban BC where gym density is high but integration is low (Fitness Avenue, 2025b).

2.2 Problem Analysis: Unmet Needs and Current Limitations

Current solutions fail to address key unmet needs, exacerbating market fragmentation and user dissatisfaction. Analysis reveals several limitations:

- **High Cumulative Costs and Inefficiency:** Users needing multi-gym access often pay \$120–\$160/month for separate subscriptions, far exceeding single-gym averages of \$40–\$90/month in urban areas (Fitness Avenue, 2025b; Yard Athletics, n.d.). This results in overpayment for underused memberships, with 45% of Canadians reporting "cost" as a top barrier to fitness (Health & Fitness Association, 2025).
- **Lack of Flexibility and Convenience:** Existing platforms like ClassPass focus on credit-based class bookings (e.g., yoga studios), with restrictions on repeat visits and expiring credits, leading to user complaints about availability and hidden fees (Trustpilot, n.d.; Yahoo Health, 2025). Wellhub (Gympass) is corporate-only, excluding individual consumers. No national platform offers unlimited-style access to traditional weight-training gyms without these constraints.
- **Administrative Burdens:** Multiple apps, physical cards, and cancellation processes (often requiring in-person visits) create hassle. Reviews show 3.5/5 average satisfaction

for competitors, with frequent mentions of "denied access" and poor support (Trustpilot, n.d.).

- **Gym Partner Challenges:** Independent gyms struggle with low off-peak occupancy (30–50% utilization) and marketing costs, while large chains dominate, leaving smaller facilities underserved (IBISWorld, 2025).

These limitations result in lower industry penetration and missed revenue opportunities, as users abandon fitness goals due to inconvenience.

2.2.1 Limitations of Existing Solutions and Identified Market Gap

Market research shows that while several digital fitness platforms exist, they do not adequately address the needs of users who require flexible access to traditional gym facilities. One prominent platform, ClassPass, operates primarily as a **credit-based marketplace for fitness classes and boutique studio sessions**. Users must book time-specific classes and spend variable credits depending on location, demand, and activity type. This model is effective for structured group workouts but is not optimized for **frequent, unscheduled gym usage**, such as independent weight training or cardio workouts.

A significant segment of fitness users prioritizes spontaneity, location flexibility, and predictable costs. These users often attend gyms based on daily schedules, commuting patterns, or travel requirements and do not wish to reserve sessions in advance. Existing platforms fail to support this behavior due to pricing complexity, credit expiration, booking constraints, and limited representation of traditional gyms. As a result, users are forced to either maintain multiple gym memberships or pay high drop-in fees, leading to inefficiency and reduced long-term engagement.

From a gym partner perspective, many independent and mid-sized gyms experience low utilization during off-peak hours but lack access to platforms that fairly monetize unused capacity. Credit-based systems often introduce revenue uncertainty and reduced control for gym operators, making them less attractive for facilities that focus on open-gym access rather than class-based programming.

This analysis reveals a clear market gap for a platform that provides **subscription-based, unscheduled access to traditional gym facilities**, with transparent pricing and simple entry mechanisms. OneFitPass Canada is designed specifically to address this gap by functioning as an access infrastructure platform rather than a class-booking marketplace. By focusing on visit-based subscriptions, QR-based check-ins, and equitable revenue sharing for gym partners, the proposed system targets an underserved user segment and complements existing class-oriented fitness platforms rather than replicating them.

The below Table summarizes the structural differences between existing fitness platforms and the proposed OneFitPass Canada solution, highlighting the identified market gap in traditional gym access.

Aspect	ClassPass	OneFitPass Canada (Proposed)
Primary Focus	Fitness classes and boutique studio sessions	Traditional gym access and open workout facilities
Access Model	Credit-based booking system	Visit-based subscription model
Booking Requirement	Advance booking required	No booking required (walk-in access)
Usage Pattern	Scheduled, time-specific sessions	Unscheduled, spontaneous gym visits
Pricing Transparency	Variable credit cost per session	Fixed monthly subscription tiers
Target User Behavior	Occasional class attendees	Frequent gym users and commuters
Gym Type Coverage	Boutique studios and wellness centers	Independent and mid-sized gyms
Entry Mechanism	Class reservation and check-in	QR-based instant check-in
Gym Partner Revenue	Credit-based payout, variable	Pay-per-visit, predictable revenue
Market Positioning	Class marketplace	Access infrastructure platform

As shown in Table, existing platforms focus primarily on scheduled, class-based experiences, whereas OneFitPass Canada targets unscheduled access to traditional gym facilities, addressing an underserved user segment.

2.3 Solution to the Identified Problem

OneFitPass Canada directly resolves these issues through a unified digital platform offering flexible, multi-gym access via a single subscription and QR-based check-in.

- **Addressing User Needs:** Tiered plans (\$49–\$159/month) provide cost savings (e.g., \$2.60–\$5 per visit at full usage) and true flexibility, allowing seamless switches between partner gyms without extra fees or apps. This reduces barriers for personas like urban

professionals and students, potentially increasing engagement by 20–30% based on similar models (ClassPass partner data, 2025).

- **Overcoming Limitations:** Unlike credit-restricted competitors, OneFitPass emphasizes unlimited options for traditional gyms, with real-time validation to prevent access issues. Gym partners benefit from pay-per-visit revenue (60–75% share), filling off-peak slots without upfront costs.
- **Broader Impact:** By integrating fragmented facilities, the platform fosters higher user retention and industry growth, aligning with post-pandemic demand for adaptable fitness solutions (Health & Fitness Association, 2025).

This solution transforms the problem into an opportunity for accessible, profitable fitness access across Canada.

3. Revenue Model

3.1 Concept

OneFitPass Canada operates on a **subscription-based platform model** with a primary revenue stream from user subscriptions, combined with a **pay-per-visit commission** structure for partner gyms. This hybrid approach aligns incentives across all parties: users pay a predictable monthly fee for flexible access, gyms receive incremental revenue for actual attendance (especially in off-peak hours), and the platform earns a sustainable margin while covering operational costs.

The core revenue streams are:

- **User Subscriptions:** Tiered monthly plans provide a set number of visits or unlimited access to the partner gym network. Pricing is designed to be competitive with single-gym memberships while offering greater value through multi-location flexibility. Example tiers (inspired by market benchmarks and adjusted for the Canadian gym landscape):
 - Basic: \$49–\$69/month for 8–12 visits (suitable for casual users).
 - Standard: \$79–\$99/month for 20–30 visits or semi-unlimited.
 - Premium: \$129–\$159/month for unlimited access

These prices position OneFitPass below the cumulative cost of 2–3 separate gym memberships (average single mid-range gym membership in Canada: \$40–\$90/month, higher in urban areas like Vancouver and Toronto at \$60–\$120) (Fitness Avenue, 2025b; Yard Athletics, n.d.).

- **Commission from Partner Gyms:** The platform takes a 25–40% commission on each verified visit (pay-per-visit payout to gyms). This is standard in multi-access fitness

platforms — for comparison, similar models in the industry often retain 20–40% after payouts, with partners receiving the remainder for filling excess capacity (Medium analysis of ClassPass model, 2025; ClassPass partner resources, 2025). Gyms benefit from no upfront marketing costs and guaranteed incremental traffic.

- **Additional Revenue Streams** (phased in after launch):
 - Premium add-ons: One-time day passes (\$15–\$25), guest access, or family plans.
 - Partnerships and advertising: In-app promotions with sports nutrition brands, apparel, or recovery services (e.g., sponsored content or affiliate commissions).
 - Data insights: Anonymized usage analytics sold to partner gyms for optimization (long-term potential).

This model ensures recurring revenue (high predictability from subscriptions) while scaling with network growth — more partners attract more users, and vice versa. It differentiates from competitors like ClassPass, which relies heavily on credit-based systems for classes/studios (often with higher effective costs and restrictions), by focusing on unlimited-style gym access without credits.

3.2 Comparison of popular single gym-subscription in Canada in 2025, Vancouver, BC

Gym Chain	Popular Plan (Access Type)	Monthly Equivalent (approx., incl. taxes)	Key Features & Notes	Source/Reference
GoodLife Fitness largest chain, 200+locations	Essential (home club)	\$35–\$45 (bi-weekly \$34.99 → ~\$76/yearly avg)	Home club only, basic equipment/classes; Ultimate: \$50–\$70 for all clubs	GoodLife Fitness (2025); Fitness Avenue (2025b)
	Ultimate (all clubs nationwide)	\$60–\$90	Multi-club access, premium classes, towel service; popular for flexibility	
Anytime Fitness (24/7, neighborhood gyms)	Standard Membership	\$50–\$75 (avg ~\$53–\$71 in Canada/BC)	24/7 access to home club + global; key fob entry, personal training options	Anytime Fitness (2025); Kent Chiro Med (2025)
Fit4Less (budget leader, GoodLife subsidiary)	4Less Card (one location)	~\$17–\$20 (\$7.99 bi-weekly)	Basic access, no-frills; Black Card: ~\$30 (\$13.99 bi-weekly) for all locations + perks	Fit4Less (2025)

Prices are approximately monthly (bi-weekly $\times 2$ + taxes $\sim 13\%$ in BC), without join/annual fees (they add \$40–\$100 in the first year).

In Vancouver/BC, prices are often 20-30% higher than the Canadian average due to urban premium (Fitness Avenue, 2025b).

Fit4Less is the cheapest for single location, but a Black Card gives multi-access for $\sim \$30/\text{month}$.

GoodLife Ultimate ($\sim \$60$ – $\$90$) is the closest competitor to the "multi—club" model but is limited only by their network (not partner halls).

3.3 Effective Cost per Visit

Effective Cost per Visit (for the user, assuming full limit usage)

- **Basic tier** (\$49–\$69/month): 8 visits $\rightarrow$ \$6.13–\$8.63 per visit 12 visits $\rightarrow$ \$4.08–\$5.75 per visit $\rightarrow$ Average range: **\$4–\$8.60** per visit
- **Standard tier** (\$79–\$99/month): 20 visits $\rightarrow$ \$3.95–\$4.95 per visit 30 visits $\rightarrow$ \$2.63–\$3.30 per visit $\rightarrow$ Average range: **\$2.60–\$5** per visit
- **Premium tier** (\$129–\$159/month, unlimited but realistic 30–45 visits/month): 30 visits $\rightarrow$ \$4.30–\$5.30 per visit 45 visits $\rightarrow$ \$2.87–\$3.53 per visit $\rightarrow$ Effective: **\$2.80–\$5.30** per visit (cheaper the more you go)

Comparison with Single Gym Memberships (2025–2026 BC/Vancouver data)

- **Fit4Less Black Card** (multi-location): $\sim \$30/\text{month}$ $\rightarrow$ unlimited in their network only
- **GoodLife Fitness Ultimate** (all clubs): \$60–\$90/month $\rightarrow$ unlimited in GoodLife network
- **Anytime Fitness**: \$50–\$75/month $\rightarrow$ home club + global access
- **Mid-range Vancouver gyms** (Steve Nash, Fitness World, etc.): \$60–\$85 single-location, \$80–\$120 multi

For 2–3 different gyms, users typically pay **\$120–\$160/month** on separate memberships.

FitPass Canada advantage:

- Access to many independent gyms (not one chain) for **\$49–\$159/month**
- At 8 visits: cheaper than one average gym (\$60–\$80) with location choice
- At 20+ visits: **\$2.60–\$5** per visit — 10–20 $\times$ cheaper than drop-in rates (\$15–\$25/day)

Platform Profitability

- User pays \$79 (Standard) $\rightarrow$ platform receives $\sim \$79/\text{month}$

- For 20 visits: pay gyms 60–75% ($\$2.60\text{--}\$3.75/\text{visit} \times 20 = \$52\text{--}\$75$) → Your commission (25–40%): **\$20–\$32** net per user (plus subscription margin)
- Unlimited tiers carry more risk, but fair usage policies + higher price cover it.

This remains highly competitive vs. ClassPass (~\$5–\$10+ effective per class/booking in Canada, with credit limits and restrictions).

3.4 Revenue Projection \$1,000,000 ARR Target

The \$1,000,000 Annual Recurring Revenue (ARR) target is realistic within 3–4 years, assuming phased rollout starting in Metro Vancouver (high gym density) and expansion to other provinces.

Key Assumptions (conservative, based on 2025 Canadian fitness data):

- Canadian gym market: ~\$5.8 billion revenue, ~9,395 facilities nationwide (IBISWorld, 2025).
- Metro Vancouver/BC focus: Hundreds of gyms (BC has ~731 facilities, with Metro Vancouver as a dense hub) (Fitness Avenue, 2025a).
- User acquisition: Start with 5,000–10,000 active subscribers in Year 1–2 via digital marketing, partnerships, and referrals.
- Average Revenue Per User (ARPU): \$80–\$100/month (blended across tiers, accounting for churn ~15–20%/year, typical for fitness apps).
- Commission margin: 30% average on visits (after gym payouts).
- Churn and growth: 20% monthly user growth in early stages, stabilizing at 10–15% YoY.

Projected Breakdown to Reach \$1M ARR:

- **Year 1:** Focus on MVP launch in BC (Vancouver area). Target 2,000–4,000 paying users by end of year. ARR ~\$200,000–\$400,000 (ARPU \$90 $\times$ 3,000 avg users $\times$ 12 months, adjusted for ramp-up).
- **Year 2:** Expand to Alberta/Ontario partnerships. Grow to 8,000–12,000 users. ARR ~\$700,000–\$900,000.
- **Year 3–4:** National scaling (10–15% partner penetration in key cities). 12,000–15,000+ subscribers + additional streams. ARR hits \$1,000,000+ (e.g., 10,000 users $\times$ \$90 ARPU = \$900,000 base + commissions/add-ons push over target).

Path to \$1M ARR Calculation Example:

- 10,000 active subscribers $\times$ \$90 average monthly subscription = \$900,000 annual subscription revenue.

- Plus ~\$100,000–\$200,000 from commissions on excess visits and add-ons = \$1M+ total ARR.

This is achievable given market size (\$5.8B industry), fragmentation (users often hold multiple memberships), and post-pandemic demand for flexible options. Risks (e.g., slower adoption) are mitigated by low customer acquisition costs through gym partnerships and viral referrals. Benchmarks from similar platforms show 20–30% revenue growth for partners, supporting mutual incentives (ClassPass partner data, 2025).

4. Feasibility Study

This feasibility study evaluates the viability of OneFitPass Canada across four key dimensions: market feasibility, technical feasibility, operational considerations, and economic feasibility. The analysis ensures that the proposed system can be successfully developed, deployed, and operated within the Canadian fitness market context.

4.1. Market Feasibility

Market feasibility assesses whether the Canadian fitness industry can support a multi-gym access platform and identifies the target audience and competitive landscape.

4.1.1. Target Market

The Canadian fitness market presents significant opportunities for OneFitPass Canada. According to IBISWorld (2025), Canada has approximately 9,395 gym, health, and fitness club businesses generating \$5.8 billion in annual revenue, with membership penetration rates between 15.5% and 21.1% of adults.

Primary Target Segments:

- **Urban Professionals (Ages 25-45):** Individuals working in dynamic environments who frequent multiple locations (home, office, transit hubs). They value flexibility and are willing to pay premium prices for convenience.
- **Students and Young Adults (Ages 18-30):** Budget-conscious users seeking variety without long-term commitments. This segment is highly responsive to digital-first platforms and referral programs.
- **Frequent Travelers and Remote Workers:** Users who work from various locations within Metro Vancouver or travel between cities. They require seamless gym access across multiple facilities.

- **Fitness Explorers:** Users interested in trying different workout environments, equipment, and class offerings before committing to a single facility.

Geographic Focus:

The initial launch targets Metro Vancouver (population 2.6 million), which contains a high density of fitness facilities and a health-conscious demographic. British Columbia has approximately 731 fitness facilities (Fitness Avenue, 2025a), with Metro Vancouver representing the largest concentration.

Market Size Estimation:

Metric	Value
Canadian Fitness Industry Revenue	\$5.8 billion (2025)
Number of Facilities (Canada)	~9,395 gyms and fitness clubs
BC Fitness Facilities	~731 facilities
Adult Gym Membership Penetration	15.5% - 21.1%
Addressable Market (Vancouver)	~400,000 - 550,000 active gym-goer

4.1.2. Competitor Analysis

The multi-gym access market in Canada is currently served by two primary platforms: Wellhub (formerly Gympass) and ClassPass. Our analysis identifies significant gaps and opportunities for differentiation.

Competitive Landscape Overview:

Aspect	Wellhub (Gympass)	ClassPass	OneFitPass (Proposed)
Access Model	Corporate-only (B2B)	Individual (B2C), credit-based	Hybrid B2C + B2B, subscription tiers
Target Focus	Employee wellness programs	Boutique studios, classes	Traditional gyms + flexibility
Pricing Model	Tiered through employer	Credits expire monthly	Transparent, price-lock guarantee

Visit Restrictions	Frequency caps on premium studios	Credit limits, late cancellation fees	Flexible home gym + variety credits
Gym Partner Relations	Payment structure disadvantages gyms	Small business owners rate 3/10	Gym-friendly revenue sharing

Identified Weaknesses in Current Platforms:

Category	Current Platform Weakness	OneFitPass Opportunity
Pricing Stability	Frequent price increases; gyms moved to higher tiers without notice	Price-lock guarantee for 12-24 months; transparent tier structure
Partnership Quality	Listed gyms sometimes no longer participate; access denied at facilities	Real-time partnership verification; quality control system
Credit Flexibility	Credits expire monthly; no rollover; sharing not allowed	Rollover credits; gift/share options; dynamic off-peak pricing
Gym Owner Experience	Poor communication; payment loopholes; unfavorable revenue split	Fair 60-75% revenue share; partnership success managers
Geographic Coverage	Broad but shallow; inconsistent quality across regions	Deep local coverage (80%+ of quality gyms) before expansion
Customer Service	Mixed reviews; issues not resolved despite multiple contacts	Proactive support; instant issue resolution; user advocacy

Market Feasibility Conclusion:

The market analysis confirms strong feasibility for OneFitPass Canada. The Canadian fitness industry is valued at \$5.8 billion with fragmented membership options. Existing competitors focus either on corporate clients (Wellhub) or boutique class bookings (ClassPass), leaving a significant gap for a consumer-focused, traditional gym access platform with transparent pricing and gym-friendly partnerships.

4.2 Technical Feasibility

Technical feasibility evaluates whether the required technology is available, mature, and suitable for the OneFitPass platform. Following Bause et al.'s (2014) framework, this assessment examines whether the technical problem can be solved under the given boundary conditions, including available resources, team capabilities, and time constraints. As Kivento (2006, as cited in Bause et al., 2014) notes, "technical feasibility means 'achievable'—this has to be proven without building the system" through defining technical options that are feasible within known requirements.

4.2.1 Technology Framework

OneFitPass will be built using established, production-ready technologies that support scalability, security, and maintainability.

Proposed Technology Stack:

Layer	Technology	Justification
Frontend (Web)	React.js with TypeScript	Industry standard; strong community; reusable components
Frontend (Mobile)	React Native	Cross-platform (iOS/Android); code reuse with web; native performance
Backend	Node.js with Express or NestJS	JavaScript ecosystem consistency; high performance; async handling
Database	PostgreSQL (primary) + Redis (caching)	ACID compliance; strong relational features; proven at scale
Cloud Infrastructure	AWS (Amazon Web Services)	Canadian data centers (Montreal, Calgary); PIPEDA compliance; auto-scaling
Payment Gateway	Stripe	PCI-DSS Level 1 certified; Canadian payment support; subscription billing
Authentication	Auth0 or AWS Cognito	OAuth 2.0/OIDC standards; MFA support; social login integration
QR Code Generation	QRCode.js + Server-side validation	Dynamic codes; time-limited tokens; fraud prevention

Technical Requirements Assessment:

- Scalability: AWS auto-scaling groups and load balancers support 10,000+ concurrent users as specified in non-functional requirements.
- Security: TLS 1.3 encryption, tokenized payments via Stripe, and role-based access control (RBAC) meet security requirements.
- Performance: Redis caching and CDN distribution ensure sub-2-second load times for core features.
- Compliance: AWS Canadian regions (ca-central-1) ensure data residency compliance with PIPEDA.

4.2.2 Membership and Check-in System

The check-in system is the core technical component enabling seamless gym access. The proposed solution uses dynamic QR codes validated in real-time.

Check-in Flow Architecture:

1. User opens the OneFitPass mobile app and navigates to the membership screen.
2. App requests a dynamic QR code from the server (valid for 5 minutes, single-use).
3. Gym staff or kiosk scans the QR code using the Partner App or integrated scanner.
4. Server validates: active subscription, visit allowance, facility partnership status, and fraud indicators.
5. Upon successful validation, check-in is recorded, and both user and gym receive confirmation.

Technical Safeguards:

- Dynamic QR Codes: Time-limited tokens prevent screenshot sharing and credential reuse.
- Geolocation Verification: Optional GPS validation ensures user proximity to the gym facility.
- Offline Fallback: Limited offline check-in capability with encrypted local storage for low-connectivity scenarios.
- Fraud Detection: ML-based anomaly detection flags suspicious patterns (e.g., rapid multi-location check-ins).

Partner Integration Options:

Integration Level	Implementation	Best For
Basic (Manual)	Staff uses Partner App on tablet/phone to scan user QR codes	Small independent gyms; minimal setup required

Standard (Kiosk)	Dedicated tablet kiosk at gym entrance with scanner	Medium facilities; self-service check-in
Advanced (API)	Direct API integration with existing gym management systems	Larger chains; automated access control integration

4.2.3 Operational Cost

Operational costs are estimated based on industry benchmarks and the technical architecture requirements for a SaaS fitness platform.

Monthly Operational Cost Breakdown (Year 1 Estimates):

Cost Category	Monthly Estimate	Annual Estimate
Cloud Infrastructure (AWS)	\$800 - \$1,500	\$9,600 - \$18,000
Payment Processing Fees (Stripe ~2.9%)	Variable (~\$2,900 at \$100K MRR)	~\$34,800 (at scale)
Third-party Services (Auth0, monitoring)	\$200 - \$500	\$2,400 - \$6,000
Customer Support Tools	\$100 - \$300	\$1,200 - \$3,600
Domain, SSL, CDN	\$50 - \$100	\$600 - \$1,200
Total Infrastructure/Operations	\$1,150 - \$2,400 (excl. processing)	\$13,800 - \$28,800 (excl. processing)

4.2.4 Technical Feasibility Conclusion

The technical feasibility analysis confirms that OneFitPass can be developed using mature, widely adopted technologies. The proposed stack (React/React Native, Node.js, PostgreSQL, AWS) is proven at scale, with strong community support and available developer talent. The QR-based check-in system addresses fraud concerns while maintaining simplicity. Initial operational costs are manageable for an early-stage startup and scale proportionally with user growth.

4.3 Estimated Financial Feasibility

Economic feasibility assesses whether the benefits of OneFitPass outweigh the costs and whether the project can achieve profitability within a reasonable timeframe. Following the DIN69901-2 standard for project management feasibility evaluation, this analysis determines whether project objectives can be achieved under the specified timeline and available resources (Bause et al., 2014). Notably, Palvia and Palvia's (1988) research found that while economic factors are important, users prioritize tangible operational benefits over intangible potential—systems must be sold on "their real costs (i.e., operational costs) and benefits" rather than development economics alone (p. 218).

4.3.1 Development Cost Estimates

Development Phase	Duration	Estimated Cost
MVP Development (Core platform, mobile app, partner portal)	4-6 months	\$80,000 - \$120,000
UI/UX Design	2-3 months	\$15,000 - \$25,000
QA Testing and Security Audit	1-2 months	\$10,000 - \$20,000
Legal (Terms, Privacy Policy, Contracts)	Ongoing	\$5,000 - \$10,000
Initial Marketing and Launch	3 months	\$10,000 - \$20,000
Total Initial Investment	6-9 months	\$120,000 - \$195,000

4.3.2 Revenue Model and Projections

Revenue is generated through subscription fees with a commission-based payout model to partner gyms.

Subscription Tiers:

Tier	Monthly Price	Visits Included	Cost Per Visit (User)
Basic	\$49 - \$69	8 - 12 visits	\$4.08 - \$8.63
Standard	\$79 - \$99	20 - 30 visits	\$2.63 - \$4.95

Premium	\$129 - \$159	Unlimited	\$2.87 - \$5.30 (at 30-45 visits)
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Revenue Projections (3-Year Path to \$1M ARR):

Metric	Year 1	Year 2	Year 3	Year 4
Active Subscribers (EOY)	2,000 - 4,000	8,000 - 12,000	12,000 - 15,000	15,000+
Avg. Revenue Per User (ARPU)	\$85/month	\$90/month	\$90/month	\$90/month
Estimated ARR	\$200K - \$400K	\$700K - \$900K	\$1.0M - \$1.3M	\$1.3M+
Partner Gyms	20 - 40	60 - 100	100 - 150	150+

4.3.3 Profitability Analysis

Unit Economics (Per User):

- Average Subscription Revenue: \$90/month
- Gym Payout (60-75% of visit value): ~\$54-\$67/month
- Payment Processing (~2.9%): ~\$2.60/month
- Infrastructure Cost (allocated): ~\$1-2/month
- Gross Margin Per User: \$20-\$32/month (22-35%)

Break-Even Analysis:

- Fixed Monthly Costs (Operations, Marketing): ~\$15,000 - \$25,000
- Variable Margin Per User: ~\$25/month
- Break-Even Subscribers: 600 - 1,000 users
- Estimated Break-Even Timeline: Month 12-18

4.3.4 Economic Feasibility Conclusion

The economic feasibility analysis demonstrates that OneFitPass Canada is financially viable. With an initial investment of \$120,000-\$195,000, the platform can achieve break-even at approximately 600-1,000 subscribers (achievable within 12-18 months). The path to \$1M ARR within 3-4 years is realistic given the market size (\$5.8 billion industry) and identified

competitive gaps. Unit economics show a sustainable gross margin of 22-35% per user, with opportunities to improve through scale and operational efficiency.

4.4 Feasibility Study Summary

Feasibility Type	Assessment	Key Findings
Market Feasibility	FEASIBLE	\$5.8B Canadian market; clear competitive gaps; identifiable target segments; existing platforms leave user and gym partner needs unmet
Technical Feasibility	FEASIBLE	Mature technology stack; proven architectures; available developer talent; manageable infrastructure costs; established security standards
Operational Feasibility	FEASIBLE	Scalable check-in system; multiple partner integration options; clear user flows; compliance requirements addressable
Economic Feasibility	FEASIBLE	Break-even at 600-1,000 users; \$1M ARR achievable in 3-4 years; sustainable unit economics; manageable initial investment
Schedule Feasibility	FEASIBLE	MVP deliverable within academic term; phased approach reduces risk; clear milestones defined

Based on this comprehensive feasibility analysis, the OneFitPass Canada project is deemed **FEASIBLE** across all dimensions. The team recommends proceeding to the Design phase with confidence in the project's viability. The identified risks are manageable with the proposed mitigation strategies, and the market opportunity justifies the investment.

5.Functional Requirements

5.1 General Platform Authentication and Security

5.1.1 Authentication

- Users can log in and log out of the system

- The system must automatically log out users after a defined period of inactivity
- Users can update their authentication credentials (e.g. reset password)
- Multi-factor authentication (MFA) for gym partners and administrators is implemented

5.1.2 Authorization & Access Control

- The system assigns a role to each account (user, gym partner, administrator) and restricts access to features based on this role (RBAC)
- The system must prevent unauthorized access attempts and block access to restricted resources

5.1.3 User & Account Functions

- Accounts can be created, modified, and deactivated
- Personal users can create an account on the platform using an email and a password
- The system must support multi-factor authentication to authenticate gym-partners and administrators

5.1.4 Security-Relevant System Behavior

- Failed login attempts must be counted and logged
- The user must be notified of failed login attempts when a registered payment method is associated with the account
- Security-relevant events shall be logged

5.2 Platform Payment and Membership System

5.2.1 Subscription Purchase and Management

- Users shall be able to activate, renew, cancel, pause, upgrade, or downgrade subscriptions
- Subscriptions with different plans and durations (weekly, monthly, or yearly contract) can be viewed and purchased
- Automatic subscription renewal unless cancelled by the user
- Automatic subscription expiration in case of failed payment

- Access to platform features and gyms is granted or restricted based on the subscription status
- Users can view their current subscription status and subscription history including start and end dates

OneFitPass-User:

- The user gets to choose different subscriptions which differentiate in weekly usage frequency and accessible gyms

Gym-Partner:

- Gym partners can select optional advertising or visibility subscription plans

5.2.2 Gym Partner Payment Processing

- Every user check-in per gym gets recorded
- Gym partners receive payouts based on the offered services, facility size, original prices for regular users, and the actual usage frequency by OneFitPass-users
- The system records each user check-in per gym
- The system shall automatically calculate and transfer the amount of payout to the gym based on selected features and analyzed user usage data at the end of the month

5.2.3 Financial Management and Reporting

- Recurring payments for active subscriptions are supported
- Invoices and payment receipts are automatically generated
- Users and gym partners shall be able to view their payment and billing history
- The system shall provide financial reports for administrators
- All payment-related and financial events are logged
- The system shall support platform-level fee, commission, and tax calculations where applicable
- Administrators can view revenue and payout summaries and export financial data
- The system handles failed payment transactions, retries, and payment adjustments

5.3 Gym User-Facing Platform

5.3.1 User Registration and Profile Management

- Users can register using required personal (i.e. full name, date of birth, address) and payment information
- The system shall validate user input and display appropriate error messages
- Users can update profile data
- Users shall be able to manage privacy settings
- Users shall be able to request account deletion

5.3.2 Gym Discovery and Check-in

- Users shall be able to search for gyms by location, opening hours, and available facilities
- Each gym shall have a detail page displaying facilities and amenities
- A digital check-in mechanism using QR codes is provided
- Check-in abuse and fraudulent usage shall be prevented

5.3.3 Visit History and Subscription Management

- Users can view their complete visit history
- The system displays remaining visit limits based on the active subscription and notifies the user, when he is close to reaching the limit

5.3.4 Gym User Dashboard

- The user dashboard shows active subscription details, recent visits, upcoming renewals, relevant notifications concerning gyms

5.4 Gym Partner-Facing Platform

5.4.1 Gym Registration and Verification

- Gym partners shall be able to register their facility on the platform
- Gym partners need to be verified to open an account, therefor they must submit the legal business name, business address, contact (optional: business registration document)

5.4.2 Facility and Schedule Management

- Facility details (opening hours, availability, equipment, amenities, courses) can be managed by gym partners

5.4.3 Partner Dashboard and Visitor Management

- Gym partners have access to a dashboard showing the number of check-ins, visit history, and payout estimates
- The system shall provide visitor statistics for operational insights

5.5 Administrator Dashboard

5.5.1 User and Platform Management

- Administrators can manage user accounts, gym partner accounts, and roles
- Administrators can configure subscription plans, pricing, and commission rates

5.5.2 Quality Assurance and Dispute Resolution

- Administrators can review disputed check-ins or payments
- Administrators shall be able to suspend or restrict accounts in case of misuse or fraud

6. Non-Functional Requirements

6.1 Usability

- The gym check-in process shall be completed in no more than three interaction steps (open app, open membership confirmation, access gym)
- Gym discovery (location, opening hours, availability) shall be accessible within three clicks
- The platform shall be designed mobile-first and be fully responsive across smartphones, tablets, and desktop devices
- Core user functions (subscribe, modify subscription, cancel subscription, view visit history) shall be available
- Error messages shall be clear and action-oriented
- The platform shall fully support English, with architectural provision for future multilingual support

6.2 Performance

- Average loading time for core features (login, check-in) shall not exceed 2 seconds under normal load
- Average loading time of gym partner search shall not exceed 4 seconds under normal load
- Check-ins shall be server-validated in under 1 second
- Backend APIs shall maintain response times of less than 500 milliseconds for standard queries
- The system shall handle peak usage periods without noticeable performance degradation

6.3 Scalability

- The system architecture shall support at least 10,000 concurrent active users
- The platform shall scale from a regional MVP (minimum viable product) to nationwide operation across Canada without fundamental architectural redesign
- The platform shall be capable of managing 1,000+ partner gyms and 50,000+ registered users

6.4 Reliability

- The platform shall achieve a minimum monthly availability of 99.5%
- Check-ins, payments, and payouts shall be transaction-safe, ensuring no data loss
- The system shall support a limited offline check-in functionality
- Automated backups of data shall be performed at least daily

6.5 Security

- The platform shall comply with Canadian data protection legislation (PIPEDA)
- Payment processing shall be handled exclusively through PCI-DSS-compliant payment providers
- User and partner accounts shall be protected using secure authentication mechanisms
- QR codes shall be dynamic to prevent misuse or credential sharing
- All data transmission shall be encrypted using TLS and HTTPS

6.6 Compatibility

- The mobile application shall be compatible with current versions of iOS and Android
- The web platform shall be compatible with major browsers (Chrome, Edge, Safari, Firefox)
- The platform shall support integration with external systems (i.e. payment gateways, analytics tools, and gym management systems)

6.7 Maintainability

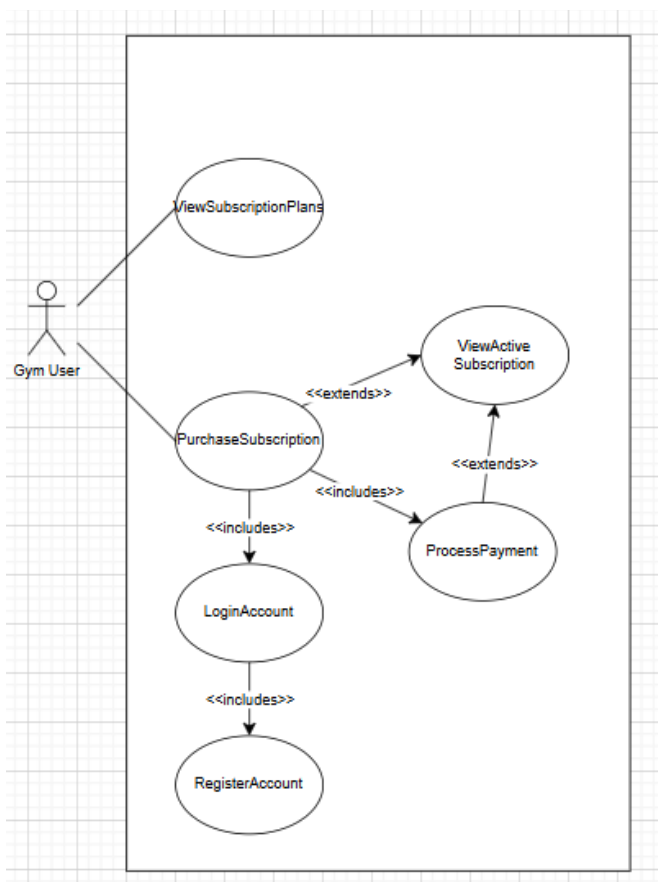
- The codebase shall be modular, well-documented, and version-controlled
- Subscription plans, pricing, and commission rates shall be configurable through the administrative role without requiring any modification to the system code.
- Monitoring and logging mechanisms shall be implemented to enable early detection of system issues
- System updates and maintenance shall be deployable with minimal service interruption

7. Use Case Diagram and Descriptions

7.1 Gym User Use Case Diagram

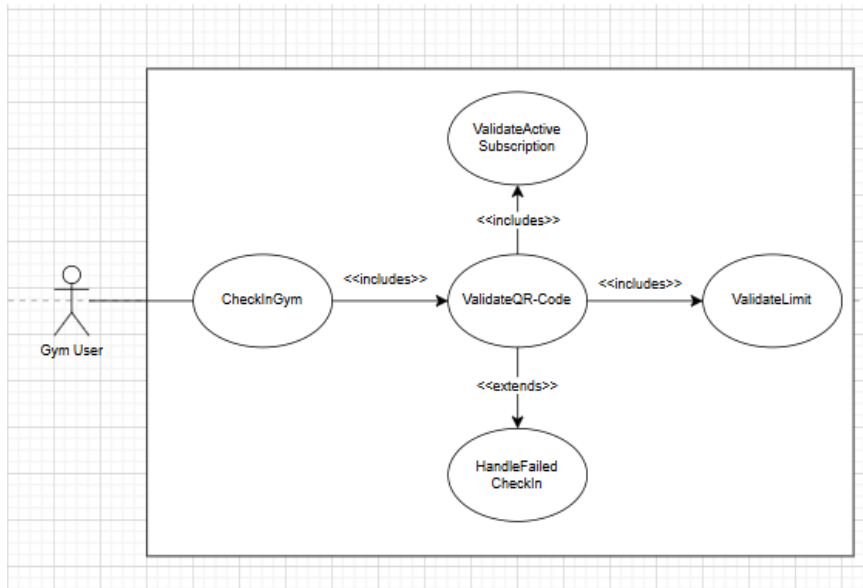
7.1.1 Purchase Subscription

This use case covers the process of viewing and purchasing a subscription plan. The user purchases a subscription, completes the payment, and receives access to partner gyms according to the selected subscription.



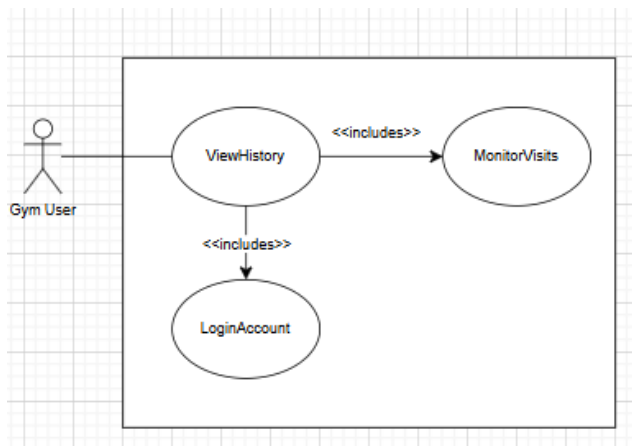
7.1.2 Check-In at Partner Gym

This use case describes how a user checks in at a partner gym using a digital QR code. The system verifies the user's subscription and visit-limits before granting gym access.



7.1.3 View Visit History

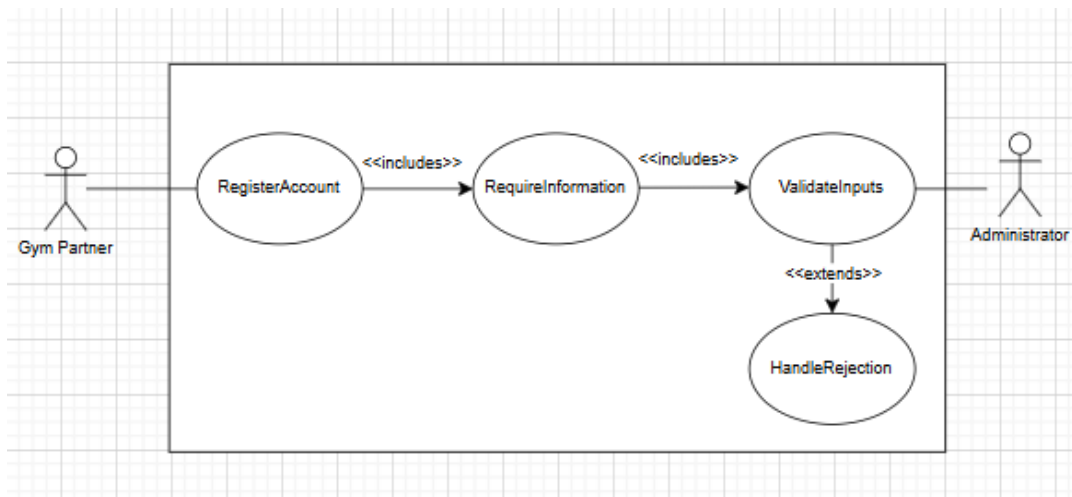
This use case allows users to review their past gym visits. The system displays a history of check-ins, enabling users to track their activity.



7.2 Gym Partner Use Case Diagram

7.2.1 Register as Partner Gym

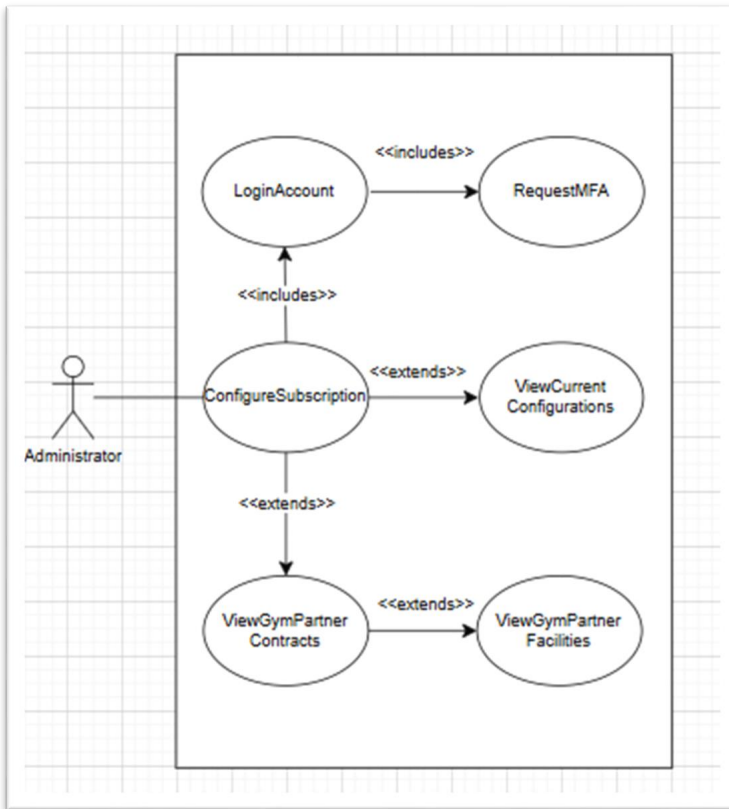
This use case explains how a gym applies to become a partner. The system stores the submitted information and sends the administrator a notification that a validation is pending.



7.3 Administrator Use Case Diagram

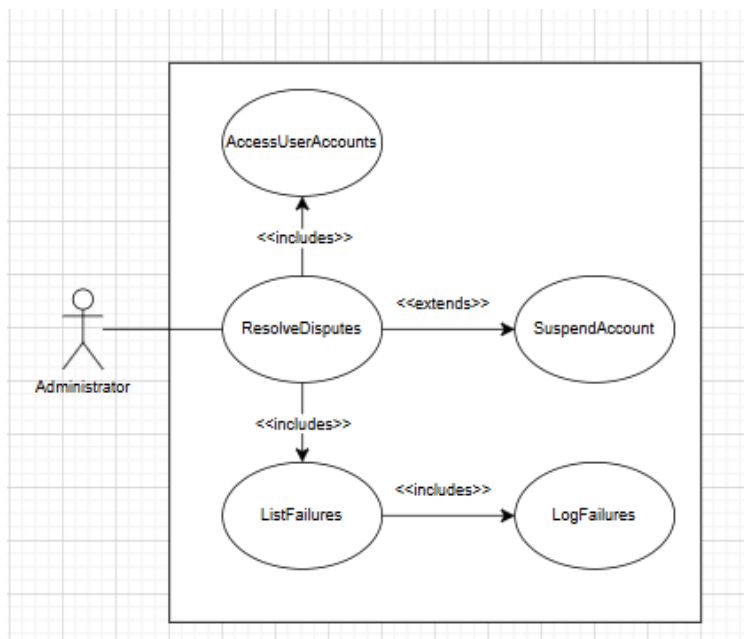
7.3.2 Configure Subscription Plans

This use case covers the management of subscription plans, pricing, and commission rates. Administrators can adjust these parameters through the system to support business and revenue requirements.



7.3.3 Resolve Disputes

This use case explains how administrators handle disputes related to check-ins or payments. The system provides relevant information and logs to support dispute resolution.



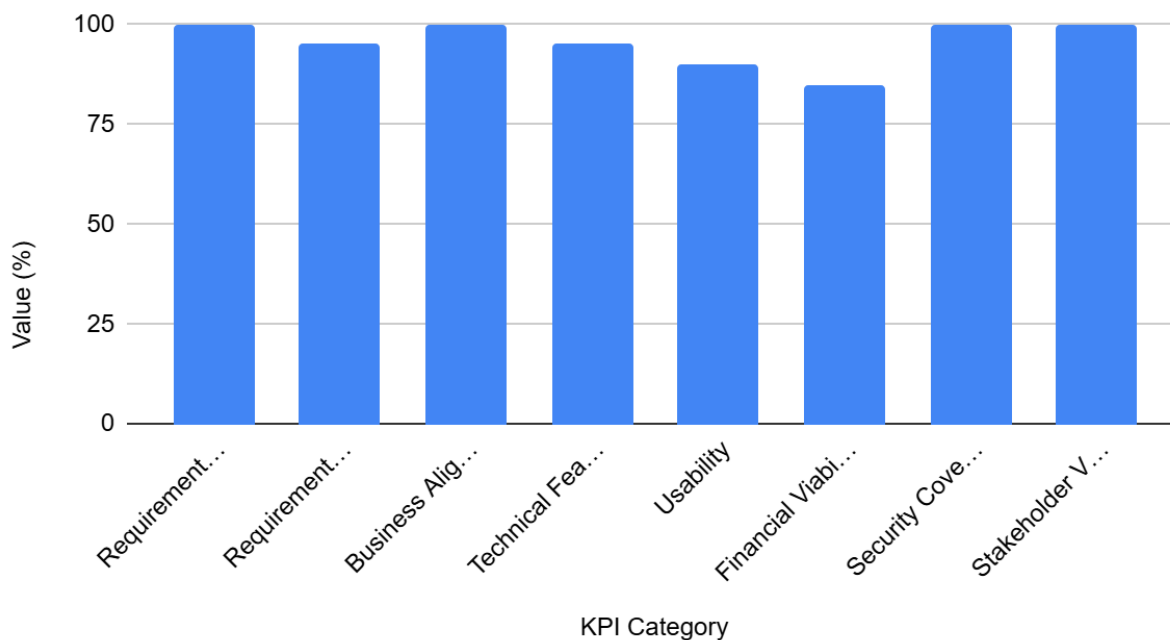
8. Key Performance Indicators (KPIs) for Requirement Analysis Phase

The Key Performance Indicators (KPIs) for the requirement analysis phase define measurable criteria to evaluate the completeness, correctness, and feasibility of the proposed OneFitPass Canada system before moving to design and implementation. These KPIs ensure that the requirements are aligned with business objectives, technically achievable, user-centered, and financially sustainable.

KPI Category	KPI Name	Metric / Measurement	Target Value	Purpose
Requirement Quality	Requirement Completeness	% of functional & non-functional requirements documented	100%	Ensures no critical system features are missing
Requirement Quality	Requirement Clarity	% of requirements with unambiguous acceptance criteria	$\geq 95\%$	Reduces misinterpretation during design & development
Business Alignment	Business Model Coverage	% of revenue-related features mapped to requirements	100%	Confirms system supports subscription & commission model
Technical Feasibility	Feasible Requirements Ratio	% of requirements implementable with current technology	$\geq 95\%$	Avoids unrealistic or risky system features
Usability	User Flow Simplicity	Avg. steps to complete core tasks (check-in, subscribe)	≤ 3 steps	Ensures smooth user experience
Financial Viability	Cost-to-Revenue Alignment	Projected margin supported by requirements	22–35%	Confirms profitability at scale

Risk & Security	Security Coverage	% of security requirements defined (auth, data, payments)	100%	Ensures compliance & fraud prevention
Stakeholder Validation	Stakeholder Approval Rate	% of team/instructor approval on requirements	100%	Confirms readiness to proceed to design

Value (%) vs. KPI Category



8.1 Requirement Quality and Completeness

Objective: Ensure that all system requirements are clearly defined and traceable.

Measures

- 100% of functional requirements documented with acceptance criteria
- 100% of non-functional requirements defined (performance, security, scalability)
- Traceability matrix linking business goals → features → use cases
- Less than 5% requirement ambiguity after peer review

Target Outcome

A Complete Software Requirement Specification with no critical gaps.

8.2 Business Model Alignment

Objective: Validate that requirements support the revenue and operational model.

Measures

- Subscription tiers mapped to user personas
- Commission structure validated against profitability model
- Cost-per-visit calculation remains within 25–40% margin
- Requirements support \$1M ARR roadmap

Target Outcome

System features directly enable subscription billing, payouts, and growth strategy.

8.3 Technical Feasibility Validation

Objective: Confirm that proposed features can be implemented with available technology.

Measures

- Proof of concept for QR check-in flow
- Payment gateway compliance for Canada
- Architecture supports 10,000+ concurrent users
- Integration feasibility with partner gyms

Target Outcome

No requirement depends on unrealistic or unavailable technology.

8.4 User Experience and Usability

Objective: Ensure the system is simple and valuable for end users.

Measures

- Check-in process ≤ 3 steps
- Gym discovery within 2 clicks

- Mobile-first responsive design
- Positive feedback from sample users

Target Outcome

High usability before UI development begins.

8.5 Financial and Operational Viability

Objective: Confirm sustainable operations.

Measures

- Projected revenue > operational cost
- Payout workflow defined
- Support effort within budget
- Fraud prevention requirements defined

Target Outcome

Platform remains profitable at scale.

8.6 Risk Mitigation and Compliance

Objective: Protect data and transactions.

Measures

- Privacy requirements aligned with Canadian laws
- Secure authentication & encryption
- Audit trail for check-ins and payments

Target Outcome

Reduced legal and security risks.

8.7 Stakeholder Validation and Sign-off

Objective: Ensure agreement from all parties.

Measures

- Team approval of SRS
- Feedback from at least 3 potential gym partners
- Instructor review acceptance

Target Outcome

Formal sign-off to proceed.

8.8 Market Validation Metrics

Objective: Verify real demand.

Measures

- Interest from ≥ 10 gyms
- Survey with ≥ 30 potential users
- Competitive comparison completed

Target Outcome

Evidence that the problem and solution are real.

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